12 tones vs 7 tones

Why the chromatic scale has 12 notes but a major or minor scale is called a 7-note scale

The chromatic scale: 12 tones

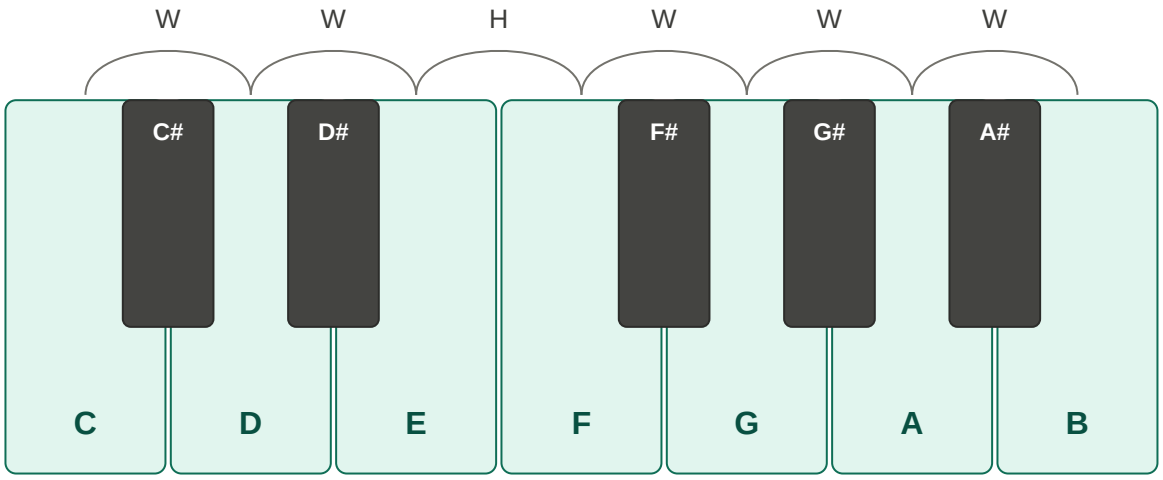
Within a single octave, there are 12 unique pitches available under the standard equal-tempered tuning system: C, C■#, D, D■#, E, F, F■#, G, G■#, A, A■#, B. This full set is the **chromatic scale** — every semitone (half-step) that exists before the pattern repeats one octave higher.

The diatonic scale: 7 tones

A major or minor scale does not use all 12 pitches. It selects 7 of them, following a fixed pattern of whole steps (W) and half steps (H). For a major scale, the pattern is:

W – W – H – W – W – W – H

The 7 notes chosen this way get the 7 letter names of the musical alphabet (A–G). The remaining 5 pitches — the sharps and flats — are called **accidentals**. They fill the gaps between certain diatonic notes and are used for chromatic movement, modulation, and color outside the home key.



Teal = 7 diatonic notes of C major Gray = 5 accidentals (chromatic notes)

One octave of a C major scale on the keyboard. The 6 arcs above the white keys show the whole/half-step pattern; the pattern closes with one more half step from B back up to the next C.

Why it matters

12 and 7 describe two different layers of the same system: 12 is the total pitch material available (the chromatic scale), while 7 is the specific subset a given key or mode draws from (the diatonic scale). Historically, the 7-note diatonic system came first; the remaining 5 in-between pitches were added later as music became more chromatic, eventually settling into the full 12-tone system used today.